

Navigators Technical Documentation

Navigators — Database Design & Entity Relationship Specification

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1. Storage Engine & Architectural Decisions

The Navigators relational data layer is built on **SQLite 3**, configured for high-concurrency embedded operation: - **Write-Ahead Logging (WAL)**: Enabled via `PRAGMA journal_mode=WAL;`, allowing concurrent readers without blocking writes. - **Foreign Key Enforcement**: Enforced via `PRAGMA foreign_keys=ON;`, guaranteeing referential integrity across all relationships. - **Synchronous Mode**: Configured to `PRAGMA synchronous=NORMAL;` for optimal balance between I/O performance and durability. - **Time Representation**: All timestamps are formatted as ISO-8601 strings in UTC (`strftime('%Y-%m-%dT%H:%M:%fZ', 'now')`).

2. Entity-Relationship Overview

```
erDiagram
    USERS ||--o{ USER_ROLES : has
    ROLES ||--o{ USER_ROLES : assigned_to
    ROLES ||--o{ ROLE_PERMISSIONS : includes
    PERMISSIONS ||--o{ ROLE_PERMISSIONS : granted_by
    USERS ||--o{ SESSIONS : establishes
    USERS ||--o{ AUTH_IDENTITYES : authenticates
    USERS ||--o{ CONTRIBUTIONS : authors
    USERS ||--o{ DATASET_SESSIONS : records
    DATASETS ||--o{ DATASET_SESSIONS : groups
    DATASETS ||--o{ TRAINING_JOBS : trains
    TRAINING_JOBS ||--o{ MODELS : produces
    MODELS ||--o{ MODEL_EVALUATIONS : evaluated_by
    CONTRIBUTIONS }o--|| PLACES : publishes_to
    USERS ||--o{ AUDIT_LOGS : triggers
```

3. Comprehensive Table Catalog (19 Tables)

3.1 Identity, Authentication & Access Control

1. users

Core user account registry. - `id` (TEXT, PK): Unique user identifier (`usr_...`). - `email` (TEXT, UNIQUE, NOT NULL): User email address. - `name` (TEXT, NOT NULL): Display name. - `status` (TEXT, NOT NULL): Account status (`active` , `suspended` , `pending_verification` , `deleted`). - `created_at` , `updated_at` , `last_login_at` (TEXT).

2. roles

Role definitions for RBAC. - `id` (TEXT, PK): Role name (`guest` , `user` , `local_contributor` , `internal_contributor` , `moderator` , `team_admin` , `super_admin`). - `name` (TEXT, UNIQUE, NOT NULL): Human-readable name. - `description` (TEXT, NOT NULL).

3. permissions

Granular atomic authorizations. - `id` (TEXT, PK): Permission string (`resource:action` , e.g., `contribution:approve`). - `resource` (TEXT, NOT NULL): Domain entity (`place` , `contribution` , `model` , `dataset`). - `action` (TEXT, NOT NULL): Operation (`create` , `read` , `update` , `delete` , `approve`). - `description` (TEXT).

4. role_permissions

Many-to-many junction mapping permissions to roles. - `role_id` (TEXT, FK → `roles.id` ON DELETE CASCADE). - `permission_id` (TEXT, FK → `permissions.id` ON DELETE CASCADE). - Primary Key: (`role_id` , `permission_id`).

5. user_roles

Many-to-many junction assigning roles to users. - `user_id` (TEXT, FK → `users.id` ON DELETE CASCADE). - `role_id` (TEXT, FK → `roles.id` ON DELETE CASCADE). - `assigned_at` (TEXT). - Primary Key: (`user_id` , `role_id`).

6. auth_identities

Multi-provider authentication credentials. - `id` (TEXT, PK): Identity record ID. - `user_id` (TEXT, FK → `users.id` ON DELETE CASCADE). - `provider` (TEXT, NOT NULL): `local_password` , `google` , `apple` . - `identifier` (TEXT, NOT NULL): Email or OAuth subject ID. - `credential_hash` (TEXT): Salted password hash.

7. sessions

Active bearer token sessions. - `id` (TEXT, PK): Unique session ID (`ses_...`). - `token_hash` (TEXT, UNIQUE, NOT NULL): SHA-256 hash of the bearer token. - `user_id` (TEXT, FK → `users.id` ON DELETE CASCADE,

NULL for guests). - `is_guest` (INTEGER, NOT NULL): 1 for anonymous sessions, 0 for authenticated. - `created_at`, `expires_at`, `revoked_at`, `last_seen_at` (TEXT). - `user_agent`, `ip_address` (TEXT).

3.2 Cartography & Community Contributions

8. `places`

Canonical points of interest rendered on the public live map. - `id` (TEXT, PK): Place identifier (`plc_...`). - `name` (TEXT, NOT NULL): Location title. - `category` (TEXT, NOT NULL): `fuel`, `hospital`, `ev_charging`, `pharmacy`, etc. - `latitude` (REAL, NOT NULL): WGS-84 latitude. - `longitude` (REAL, NOT NULL): WGS-84 longitude. - `address`, `phone`, `operating_hours` (TEXT). - `verified` (INTEGER, DEFAULT 0). - `status` (TEXT, NOT NULL): `active`, `deprecated`, `deleted`. - `source` (TEXT, NOT NULL): `canonical`, `community_contribution`. - `version` (INTEGER, DEFAULT 1).

9. `contributions`

Crowdsourced community submissions with formal lifecycle state machine. - `id` (TEXT, PK): Contribution ID (`contrib_...`). - `owner_id` (TEXT, NOT NULL): Author user ID. - `resource_type` (TEXT, NOT NULL): Target entity type (`place`, `road`). - `status` (TEXT, NOT NULL): `draft`, `submitted`, `pending_review`, `approved`, `published`, `rejected`, `changes_requested`, `withdrawn`. - `title` (TEXT, NOT NULL). - `data_json` (TEXT, NOT NULL): Proposed place attributes in JSON format. - `reviewed_by` (TEXT, FK → `users.id`). - `review_notes` (TEXT). - `target_resource_id` (TEXT, FK → `places.id`).

3.3 Telemetry, Machine Learning & Model Registry

10. `datasets`

Curated collections of driving sessions. - `id` (TEXT, PK): Dataset ID (`ds_...`). - `name` (TEXT, UNIQUE, NOT NULL). - `version` (TEXT, NOT NULL). - `split_json` (TEXT): Train/Validation/Test session mapping.

11. `dataset_sessions`

Individual driving sessions recorded by mobile devices. - `id` (TEXT, PK): Session ID (`dss_...`). - `dataset_id` (TEXT, FK → `datasets.id`). - `contributor_id` (TEXT, NOT NULL). - `activity_type` (TEXT, NOT NULL): `driving`, `walking`. - `device_model` (TEXT). - `duration_seconds` (REAL, NOT NULL). - `consent_verified` (INTEGER, DEFAULT 0). - `validation_status` (TEXT): `pending`, `validated`, `rejected`.

12. `training_jobs`

Asynchronous neural network training records. - `id` (TEXT, PK): Job ID (`job_...`). - `model_architecture` (TEXT, NOT NULL): `tcn`, `lstm`. - `dataset_id` (TEXT, FK → `datasets.id`). - `status` (TEXT): `queued`, `running`, `completed`, `failed`. - `hyperparameters_json`, `metrics_json` (TEXT).

13. `models`

Central registry of trained neural checkpoints. - `id` (TEXT, PK): Model ID (`mdl_...`). - `name` (TEXT, NOT NULL). - `version` (TEXT, NOT NULL). - `architecture` (TEXT, NOT NULL). - `checkpoint_path`, `onnx_path`

(TEXT, NOT NULL). - `status` (TEXT, NOT NULL): `candidate`, `approved`, `production`, `archived`, `rejected`. - `is_production` (INTEGER, DEFAULT 0).

14. `model_evaluations`

Benchmark evaluations associated with model candidates. - `id` (TEXT, PK): Evaluation ID (`eval_...`). - `model_id` (TEXT, FK → `models.id`). - `dataset_session_id` (TEXT, FK → `dataset_sessions.id`). - `ate_rmse`, `velocity_rmse`, `max_drift_m` (REAL).

3.4 Governance, Incidents & Operations

15. `internal_access_requests`

User applications to become Internal Contributors. - `id` (TEXT, PK): Request ID (`req_...`). - `user_id` (TEXT, FK → `users.id`). - `status` (TEXT): `pending`, `approved`, `rejected`. - `justification` (TEXT, NOT NULL). - `reviewed_by` (TEXT).

16. `incident_reports`

User-reported road obstructions, accidents, and hazards. - `id` (TEXT, PK): Report ID (`rep_...`). - `reporter_id` (TEXT, NOT NULL). - `category` (TEXT, NOT NULL): `accident`, `road_closure`, `hazard`. - `latitude`, `longitude` (REAL, NOT NULL). - `status` (TEXT): `active`, `resolved`, `false_report`.

17. `sos_alerts`

Emergency vehicular distress signals. - `id` (TEXT, PK): SOS ID (`sos_...`). - `user_id` (TEXT, NOT NULL). - `latitude`, `longitude` (REAL, NOT NULL). - `status` (TEXT): `triggered`, `acknowledged`, `resolved`, `cancelled`.

18. `audit_logs`

Immutable compliance and security ledger. - `id` (TEXT, PK): Audit ID (`aud_...`). - `actor_id` (TEXT): User ID triggering action. - `action` (TEXT, NOT NULL): e.g., `model:promoted`, `role:assigned`. - `resource_type`, `resource_id` (TEXT, NOT NULL). - `old_state`, `new_state` (TEXT). - `metadata_json` (TEXT). - `timestamp` (TEXT, DEFAULT now).

19. `sync_queue`

Idempotent server-side replay buffer for offline actions. - `id` (TEXT, PK): Sync ID (`sync_...`). - `client_sync_id` (TEXT, UNIQUE, NOT NULL): Client-generated UUIDv4 idempotency key. - `user_id` (TEXT, NOT NULL). - `action_type` (TEXT, NOT NULL). - `payload_json` (TEXT, NOT NULL). - `processed_at` (TEXT).

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